



DTE/DCE RJ45 cables?



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Author

Message

bugOnLine99

 **Posted:** 01 Aug 2012 17:58

 offline

Joined: 06 Sep 2009
 02:09
Posts: 106
Post subject: DTE/DCE RJ45 cables?

Hello,

I am a newbie when it comes to networking as I have never really worked with computer networks before so I am really green at this. About a few days ago I decided to try the Ethernet demo for PIC32. Not as simple as I thought. 😊

As I was trying to learn a little about networking and stumbled ove this web site which explains the difference between Staright through and crossed-over RJ45 Ethernet cables:

<http://www.iplocation.net/tools/rj45-wiring.php>

At my office I have my PC connected to a linksys router. According to the website listed above, the type of cable connecting my PC and my router should be a cross-over RJ45 type. To my surprise, when I looked at my cable it was a straight through type AND has always worked. So I decided to give linksys a call and asked them as to why networking articles on the web are stating that cross-over cables should be used when connecting two identical equipements (ex: DTE to DTE OR DCE to DCE), but when their Linksys router (DTE) is connected to my PC (DTE) it still works with a straight through cable.

The answer: It turns out that their routers have intelligent ports which can adapt to either type of RJ45 cable... so it really doesn't matter if I use a stright-through or cross-over cable to connect my computer to my router. I should of known about this !!! 😊

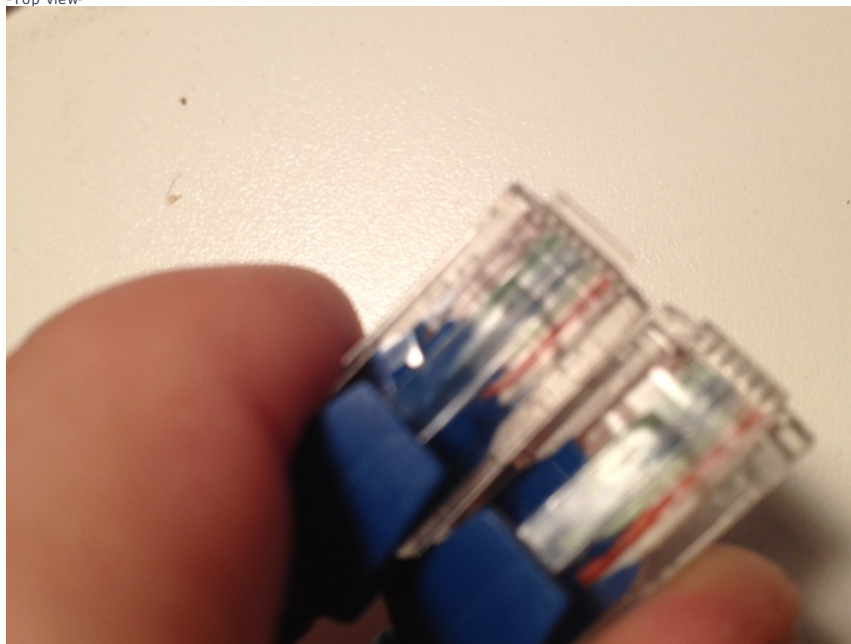
So my first question is that I was wondering if the web server on the Multimedia board for PIC 32 is a DTE? I think it would be but I am not 100% sure.

My next question is, since my router has intelligent ports, is it okay to connect a straight through RJ45 cable from my router to the multimedia board for PIC32? I can't get the demo to work and I just want to eliminate the probability that its not the cable. Please view pics for details on the cable I am using!

All help appreciated.
 r

Attachments:

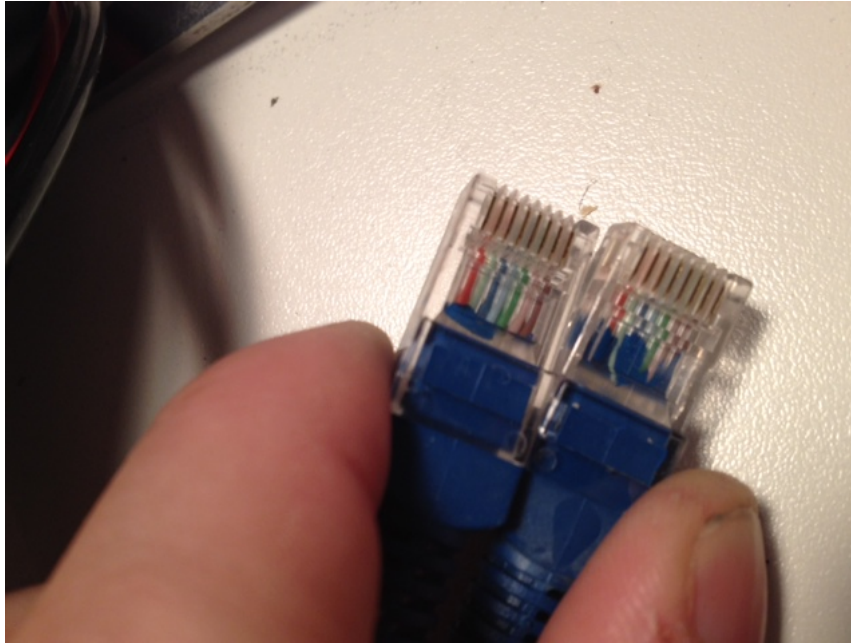
File comment: This is the cable I am using from the router to the multimedia board
 -Top view-



RJ45Top.JPG [74.55 KiB | Viewed 3297 times]

File comment: This is the cable I am using from the router to the multimedia board

-Bot view-



RJ45Bot.JPG [92.45 KiB | Viewed 3297 times]

[Top](#)[profile](#)**LGR****Post subject:** Re: DTE/DCE RJ45 cables?**Posted:** 02 Aug 2012 01:17**offline**

Go to the page where they sell the board. The schematic diagram is available in PDF.

Joined: 23 Sep 2004
20:07
Posts: 3193

If you know what you're doing, you're not learning anything.

[Top](#)[profile](#)**dejan.odabasic****Post subject:** Re: DTE/DCE RJ45 cables?**Posted:** 02 Aug 2012 08:53**offline**

mikroElektronika team



Hello,

bugOnLine99 wrote:

but when their Linksys router (DTE) is connected to my PC (DTE) it still works with a straight through cable.

Joined: 30 Apr 2012 14:20
Posts: 2416

This is wrong assumption, Linksys router is not DTE, it's DCE.

You have probably misunderstood next sentence:

Quote:

The straight-through cables are used when connecting Data Terminating Equipment (DTE) to Data Communications Equipment (DCE), such as computers and routers to modems (gateways) or hubs (Ethernet Switches)

Data Terminating Equipment (DTE) - PC clients / servers

Data Communications Equipment (DCE) - routers / gateways / hubs / Ethernet Switches

MultiMedia Board for PIC32MX7 is a DTE and if you are connecting it to the router you should use straight-through cable and cross-over cable if you want to connect MMBforPIC32MX7 directly to PC.

It's important that you set PIC32 network parameters according your router network setup.

Code: [Select all](#)

```
// network parameters
char myMacAddr[6] = {0x00, 0x14, 0xA5, 0x76, 0x19, 0x3f}; // my MAC address
char myIpAddr[4] = {192, 168, 20, 60}; // my IP address
char gwIpAddr[4] = {192, 168, 20, 6}; // gateway IP address
char dnsIpAddr[4] = {192, 168, 20, 1}; // dns IP address
char ipMask[4] = {255, 255, 255, 0}; // subnet mask
// end network parameters
```

Best regards.

[Top](#)[profile](#)

offline

Joined: 06 Sep 2009
02:09
Posts: 106

Hello dejan.odabasic,

Quote:

You have probably misunderstood next sentence:

Quote:

The straight-through cables are used when connecting Data Terminating Equipment (DTE) to Data Communications Equipment (DCE), such as computers and routers to modems (gateways) or hubs (Ethernet Switches)

I find this phrase a little misleading, but I like it better the way you explain it. 😊

My set up is as follows:

My PC and the multimedia board are both connected to the same linksys router each with a staright-through cable.

DTE to DCE = straight- through!

As for the code sample I don't really know what to put as a MAC address?? Here in the sample below... Note that I just put '01' everywhere. I was wondering, how do I know what the MAC address of the multi media board is?

Anyways, here is one of the many code variations I tried without success:

Code: Select all

```
#include "Ethernet_Demo_objects.h"
extern void InitPHY Pins();
extern char myMacAddr[6] = {0x01, 0x01, 0x01, 0x01, 0x01, 0x01};    //<<< ???
extern char myIpAddr[4] = {192, 168, 1, 133};    // Multimedia board's IP address I chose this IP address

extern unsigned char gwIpAddr[4] = {64,230,197,106};                // Default gateway in my router
extern unsigned char ipMask[4] = {255,255,255,255};                // Subnet mask in my router
extern unsigned char dnsIpAddr[4] = {207,164,234,129};              // 1st DNS addr in my router
unsigned char v;

void main() {
    Start_TP();
    //Ethernet_Enable(_Ethernet_MULTICAST); // _Ethernet_BROADCAST);
    //v = Ethernet_initDHCP(5);
    InitPHY Pins();
    Ethernet_Init(myMacAddr, myIpAddr, _ETHERNET_AUTO_NEGOTIATION & _ETHERNET_DEFAULT_MAC);

    while (1) {
        Check_TP();
        Ethernet_doPacket();
    }
}
```

When I go to the chrome browser and enter "192.168.1.133" in the address bar, the browser can't find the multimedia board on the network 😊

I tried to ping this address from my PC and the address is never found? Please view the information in my router shown in the attachment below.

Even though I read up on networking everyday, I still seem to have dificulties getting to know everything there is to know about networking...

I really appreciate your assistance and help.

regards
rob

Attachments:

LINKSYS®
A Division of Cisco Systems, Inc.

Firmware Version: v3.37.7

Status

Wireless-G Broadband Router with SpeedBooster

WRT54GS

Setup

Wireless

Security

Access Restrictions

Applications & Gaming

Administration

Status

Router

Local Network

Wireless

Router Information

Firmware Version: v3.37.7, Apr. 25, 2005

Current Time: Thu, 02 Aug 2012 20:18:22

MAC Address: 00:13:10:A1:89:03

Router Name: WRT54GS

Host Name:

Domain Name:

Internet

Configuration Type

Login Type: PPPoE

Login Status: Connected

Disconnect

IP Address: 70.52.56.126

Subnet Mask: 255.255.255.255

Default Gateway: 64.230.197.106

DNS 1: 207.164.234.129

DNS 2: 207.164.234.193

DNS 3:

MTU: 1492

Firmware Version. This is the Router's current firmware.

Current Time. This shows the time, as you set on the Setup Tab.

MAC Address. This is the Router's MAC Address, as seen by your ISP.

Router Name. This is the specific name for the Router, which you set on the Setup Tab.

More...

Configuration Type. This shows the information required by your ISP for connection to the Internet. This information was entered on the Setup Tab. You can Connect or Disconnect your connection here by clicking on that button.

More...

Refresh

CISCO SYSTEMS

lyinksys.jpg [129.41 KiB | Viewed 3267 times]

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dejan.odabasic

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 03 Aug 2012 14:48

mikroElektronika team

Joined: 30 Apr 2012 14:20
Posts: 2416

Hello,

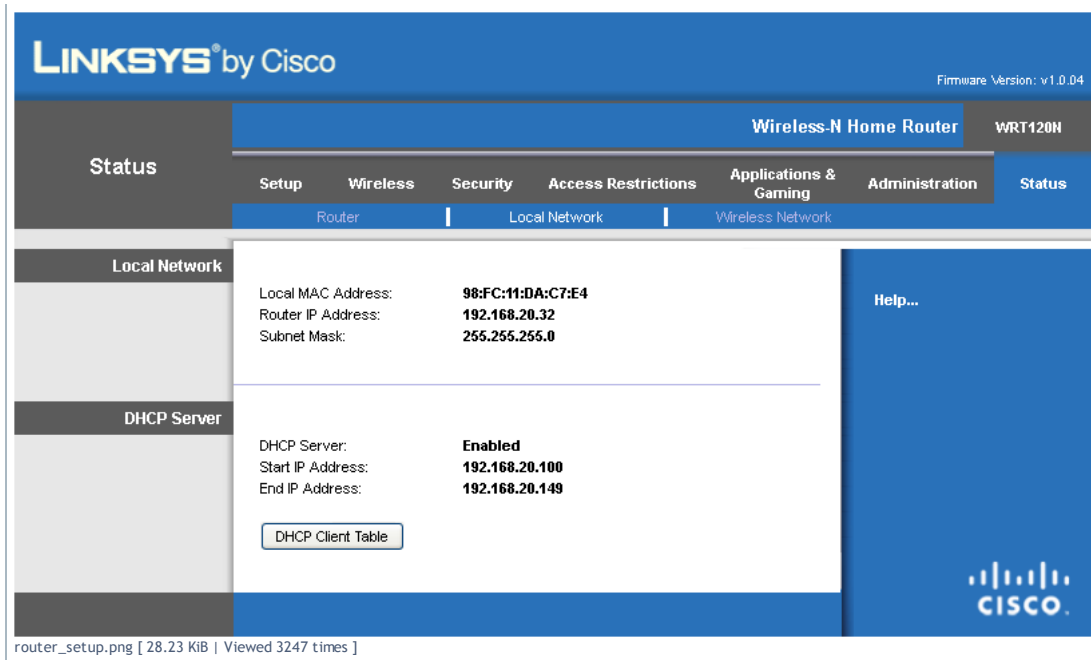
you shouldn't change the default MAC address from example project.
Only if you know that some other device in your network have the same MAC address, which is highly unlikely, then you need change it. You can use one of many [online tools](#) for generating proper MAC address.

You were looking at the wrong network setup page (Status/Router). That page shows the setup given from your internet provider.
You need to look at you local network settings (Status/Local Network) or go to (Setup/Basic setup).
Here is the example of Linksys router setup, which should not be that much different from yours.

Attachment:

<http://www.mikroe.com/forum/viewtopic.php?f=164&t=49793&hilite=RJ45>

4/16



Based on settings you see on the picture, you should set Gateway IP address to be the same as IP address of router. (192.168.20.32)
If you Subnet mask is the same (255.255.255.0) then you can chose IP address for MMB32MX7 from range (192.168.20.1 - 192.168.20.254) excluding all taken addresses by other hosts in your network.

You can find DNS IP address in you PC network adapter setup (use the same one).

When you open you browser you should enter the IP address that you set for MMB32MX7 board.

Best regards.

Top

bugOnLine99

[profile](#)

Post subject: Re: DTE/DCE RJ45 cables?

Hello dejan.odabasic,

Your help is very appreciated.

Okay,

When I sarted this demo, I somehow downloaded an Ethernet demo code with 'myMacAddr[]' being set to a particular MAC address... something like this:

```
extern char myMacAddr[6] = {0x00, 0x21, 0x97, 0x35, 0x9B, 0xE1};
```

However, by moving code around while doing tests, the original mac address was changed. I tried to download the MIKROELEKTRONIKA Ethernet Demo to tr file, but I don't see any demos with that predefined mac address anymore. All I see in every demo is:

```
// ...
extern char myMacAddr[6]; // <<< no predefined address anymore here ???
// ...
```

I tried downloading the Ethernet demo from your CD, from your website and from installation directories and nothing. The Ethernet demo with the predefin reason, the Mac address in the code sample below might not be 100% correct. Unless you can find it somewhere in MKE's samples... it would be great!

Other than that, I have made the modifications you listed... here is the code:

Code: Select all

```
#include "Ethernet_Demo_objects.h"
extern void InitPHY Pins();
extern char myMacAddr[6] = {0x00, 0x21, 0x97, 0x35, 0x9B, 0xE1}; // <<< Not 100% sure
extern char myIpAddr[4] = {192, 168, 1, 133}; // Chosen multimedia board's IP address

extern unsigned char gwIpAddr[4] = {192, 168, 1, 1}; // Same as router's IP address
extern unsigned char ipMask[4] = {255, 255, 255, 0}; // Subnet mask in my router
extern unsigned char dnsIpAddr[4] = {207, 164, 234, 129}; // The same one as my PC's DNS

void main() {
    Start_TP();
    InitPHY Pins();
    Ethernet_Init(myMacAddr, myIpAddr, _ETHERNET_AUTO_NEGOTIATION & _ETHERNET_DEFAULT_MAC);
```

```

while (1) {
    Check_TP();
    Ethernet_doPacket();
}
}

```

As for the attachments below:

Is the ipconfig /all report of my PC. Also I have included the correct router status page.

I tried to enter "192.168.1.133 and I am still not able to access the web page in the multimedi board 😞

The cable is a straight through... as shown in the pictures of this inital thread!!!

-Questions:

1) Do I need the following lines of code in my sample:

```
Ethernet_Enable(_Ethernet_MULTICAST); //_Ethernet_BROADCAST);
```

```
v = Ethernet_initDHCP(5);
```

2) My router is connected to a DSL modem... That doesn't interfere with our settings ... right?

3) A year ago my router was full and we needed to expand the number of ports. So we added a switch with four (4) more ports. So basically, right now, my router is trying this is by unplugging the cable going from the router to the switch (thus freeing up a port on the router) and replacing it with the multimedia board automatically?

Discouraged 😞

r

Attachments:

```

C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Robert>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : ROBERT
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . :
    Description . . . . . : NVIDIA nForce Networking Controller
    Physical Address. . . . . : 00-21-97-35-9B-E1
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::1817:7328:4e20:a35c%11(Preferred)
    IPv4 Address. . . . . : 192.168.1.102(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : August-03-12 1:56:07 PM
    Lease Expires . . . . . : August-04-12 1:56:07 PM
    Default Gateway . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    DHCPv6 IAID . . . . . : 234889623
    DHCPv6 Client DUID. . . . . : 00-01-00-01-13-71-2D-D5-00-21-97-35-9B-E1
    DNS Servers . . . . . : 207.164.234.129
                          207.164.234.193
    NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter Local Area Connection* 9:

    Connection-specific DNS Suffix . :
    Description . . . . . : Teredo Tunneling Pseudo-Interface
    Physical Address. . . . . : 00-00-00-00-00-00-E0
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2001:0:4137:9e76:34c1:2d5c:3f57:fe99(Pref
    Link-local IPv6 Address . . . . . : fe80::34c1:2d5c:3f57:fe99%12(Preferred)
    Default Gateway . . . . . :
    NetBIOS over Tcpip. . . . . : Disabled

Tunnel adapter isatap.{76B4BFC3-03DF-4E86-BE04-AD251EC5EBD2}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft ISATAP Adapter #2
    Physical Address. . . . . : 00-00-00-00-00-00-E0
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes

C:\Users\Robert>

```

Ipconfig_all.jpg [201 KiB | Viewed 3241 times]

The screenshot shows a web browser window with the address bar displaying `http://192.168.1.1/Status_Lan.asp`. The browser's address bar includes navigation buttons (back, forward, stop, reload) and a search icon. Below the address bar, there are search engines (Google, Ad-Aware, Blekko) and a search bar. The main content area is the Linksys router's status page. The top header is purple with the Linksys logo and the text "A Division of Cisco Systems, Inc.". Below the header, there is a navigation menu with tabs: "Status", "Setup", "Wireless", "Security", "Access Restrictions", and "Applications & Gaming". The "Status" tab is selected, and within it, the "Local Network" sub-tab is active. The "Local Network" section displays the following information:

MAC Address:	00:13:10:A1:89:02
IP Address:	192.168.1.1
Subnet Mask:	255.255.255.0
DHCP Server:	Enabled
Start IP Address:	192.168.1.100
End IP Address:	192.168.1.149

Below this information, there is a button labeled "DHCP Clients Table". At the bottom right of the page, there is a "Refresh" button. The footer of the page shows the filename "LocalNetwork.png" and its size "209.04 KiB" along with the view count "Viewed 3241 times".

[Top](#) [profile](#)**dejan.odabasic****Post subject:** Re: DTE/DCE RJ45 cables?**Posted:** 06 Aug 2012 10:30 offline

mikroElektronika team

**Joined:** 30 Apr 2012 14:20
Posts: 2416

Hello,

Your setting seems to be ok.
Could you please try to program attached project.

Attachment:

 [Ethernet Demo.zip](#) [63.41 KiB]
Downloaded 298 times

Best regards.

[Top](#) [profile](#)**bugOnLine99****Post subject:** Re: DTE/DCE RJ45 cables?**Posted:** 06 Aug 2012 18:29 offline**Joined:** 06 Sep 2009
02:09
Posts: 106

Hello dejan.odabasic,

Unfortunately, it doesn't work. I have tried your project as it was without success. Also I have used your project and modified the code to:

Code: [Select all](#)

```
#include "Ethernet_Demo_objects.h"
extern void InitPHY Pins();
extern char myMacAddr[6];
extern char myIpAddr[4] = {192, 168, 1, 133}; // <<< I guess project will set default MAC!!!
// Chosen multimedia board's IP address

extern unsigned char gwIpAddr[4] = {192,168,1,1}; // Same as router's IP address
extern unsigned char ipMask[4] = {255,255,255,0}; // Subnet mask in my router
extern unsigned char dnsIpAddr[4] = {207,164,234,129}; // The same one as my PC's DNS

void main() {
    Start_TP();
    InitPHY Pins();
    Ethernet_Init(myMacAddr, myIpAddr, _ETHERNET_AUTO_NEGOTIATION & _ETHERNET_DEFAULT_MAC);

    while (1) {
        Check_TP();
        Ethernet_doPacket();
    }
}
```


The above did not work! 😞

One thing I noticed though, is when I tried to run the Ethernet project in debug mode and for some reason I cannot step through the following line:

```
Ethernet_Init(myMacAddr, myIpAddr, _ETHERNET_AUTO_NEGOTIATION & _ETHERNET_DEFAULT_MAC);
```

In other words, the code gets stuck at this line of code ?? 😞

Also, there is one strange thing!! All MKE Ethernet demos all have the following errors when compiled:

```
....
0 122 Compilation Started ethernet_demo_objects.h
98 428 Nonexistent field 'color' Ethernet_Demo_events_code.c
99 428 Nonexistent field 'color' Ethernet_Demo_events_code.c
101 428 Nonexistent field 'color' Ethernet_Demo_events_code.c
...
0 102 Finished (with errors): 06 Aug 2012, 13:12:13 Ethernet_Demo.mcp32
```

So for now I comment out lines 98 to 102 in the ethernet_demo_events_code.c file. I believe this doesn't make a difference since this is only responsible for coloring the buttons on the LCD of the multimedia board. Somehow since the beginning, I always had this problem....and yes the TFT and the TFT_TouchPanel libraries are checked off in the library manager. It seems as though the project doesn't see the definitions of colors such as 'CL_WHITE'. I think... to get rid of these errors I would have to re-create the Ethernet demo starting from Visual TFT and then load it into MikroC compiler and then compile the whole project.

But for now I don't mind these errors... what is important is to get the communications to work for the Ethernet demo 😊

I am trying different things here with the Ethernet_Init(...) function but it always gets stuck there. So it is useless for me to go to the browser and try to enter the IP in the address bar. The issue seems to be with the Ethernet_Init(...) function 😞

Its a little weird as to why this would happen...

anyways I am standing by... thanks for your assistance... it is very kind of you to follow up 😊

regards
robert

Top

 profile

dejan.odabasic

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 07 Aug 2012 12:22

offline

Hello,

mikroElektronika team



You shouldn't get any error trying to compile the attached project. What is the version of compiler that you are using? Check for the latest version of compiler: [http://www.mikroe.com/eng/products/view ... for-pic32/](http://www.mikroe.com/eng/products/view...for-pic32/)

Also there was no need for you to edit the code.

I already set the network addresses in the Ethernet_Demo_events_code.c file.

Did you try to compile and program attached code without modification?

Best regards.

Joined: 30 Apr 2012 14:20
Posts: 2416

Top

 profile

bugOnLine99

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 07 Aug 2012 18:45

offline

Eureka!!!!

Wow! it works... 😊

Please view the Web server page I see on my PC browser at attachment.

Okay... ooooofff I am full questions 😊😊😊 LOL!!!!

>>Also there was no need for you to edit the code.

>>I already set the network addresses in the Ethernet_Demo_events_code.c file.

Sincerely, yesterday I had tried it without modifying your code. And it didn't work...

I came in this morning and I changed the RJ45 cable tried it and didn't work. I put back the original cable and re-tried your code ... and all of the sudden it worked. 😊

QUESTIONS:

1) In my router's status page, the start IP address is 192.168.1.100 and the end IP address is 192.168.1.149. Since this is the router's applicable IP range, why is it that the router is able to see the IP address of 192.168.1.60 which is out of range according to the router??

2) In the google chrome browser I enter the following address to access the web server on the multimedia board:
"192.168.1.60"

Would this also work from a smart phone? Ex: iPhone...

3) Does MKE have any detailed documentation on how to use all the other Ethernet commands?

4) I notice the following XML code in the "Ethernet_Demo_ebents_code.c" file:

```
Code: Select all

const code char *indexPage = // Change the IP address of the page to be refreshed
"<meta http-equiv=\"refresh\" content=\"3;url=http://192.168.1.60\">\n
<HTML><HEAD></HEAD><BODY>\n
<h1>PIC32 Ethernet Mini Web Server</h1>\n
<a href=/>Reload</a>\n
<script src=/s></script>\n
<table><tr><td valign=top>\n
</td><td>\n
" ;

const code char *indexPage2 = "</td><td>\n
<table border=1 style=\"font-size:20px ;font-family: terminal ;\">\n
<tr><th colspan=3>LED</th></tr>\n
<script>\n
var str,i;\n
str=\"\";\n
for(i=0;i<8;i++)\n
{str+=\"<tr><td bgcolor=yellow>LED #\"+i+\"</td>\";\n
if(PORTD&(1<i)) {str+=\"<td bgcolor=red>ON\";\n
else {str+=\"<td bgcolor=#cccccc>OFF\";\n
str+=\"</td><td><a href=/t\"+i+\">Toggle</a></td></tr>\";\n
document.write(str) ;\n
</script>\n
</table></td></tr></table>\n
This is HTTP request #<script>document.write(REQ)</script></BODY></HTML>\n
" ;
```

If I wanted to create my own web page can I modify/re-code the XML code above?

5) Can I replace the code in the "Ethernet_UserUDP()" and the " Ethernet_UserTCP()" functions with my own code in order to create some of my own control logic.

6) Does MKE have any tutorial kits for sale which would explain simple code samples for creating our own web pages in the MCU?

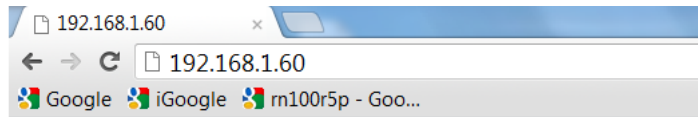
7) I would like to create my own web pages along with discrete and analog/numerical control. For example, one can set the heating set point temperature from the web page. What documentation/kits/books/articles would you suggest I read to acquire more knowledge with Ethernet technology for embedded systems.

☺ I was wondering, does the webserver in the multimedia board require the DHCP to be enabled. Where is it enabled in your Ethernet sample?

dejan.odabasic... I sincerely thank you for your help 😊

regards
roberto

Attachments:



PIC32 Ethernet Mini Web Server

[Reload](#)

LED		
LED #0	OFF	Toggle
LED #1	OFF	Toggle
LED #2	OFF	Toggle
LED #3	OFF	Toggle
LED #4	OFF	Toggle
LED #5	OFF	Toggle
LED #6	OFF	Toggle
LED #7	OFF	Toggle

This is HTTP request #32

PIC32WebServer.png [38.37 KiB | Viewed 3196 times]

[Top](#)

[profile](#)

dejan.odabasic

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 08 Aug 2012 10:42

[offline](#)

mikroElektronika team



Joined: 30 Apr 2012 14:20
Posts: 2416

Hello,

First of all I'm sorry I didn't mention that IP address is set to 192.168.1.60. 😊

I'm really glad I could help you. 😊 Also I'll try my best to answer all of you questions. 😊

bugOnLine99 wrote:

1) In my router's status page, the start IP address is 192.168.1.100 and the end IP address is 192.168.1.149. Since this is the router's applicable IP range, why is it that the router is able to see the IP address of 192.168.1.60 which is out of range according to the router??

IP range that you see on your status page is the range from which your router DHCP server will assign IP address to hosts which are expecting the address by DHCP.

You can edit that range on page (Setup/Basic setup).

Important thing is that your router IP is 192.168.1.1, and you subnet mask is 255.255.255.0 which defines that all host in range 192.168.1.1 - 192.168.1.254 are in the same network (visible to each other).

You can check this, using one of many online subnet calculators.

bugOnLine99 wrote:

2) In the google chrome browser I enter the following address to access the web server on the multimedia board:
"192.168.1.60"

Would this also work from a smart phone? Ex: iPhone...

Yes it will if you connect your phone over WiFi to your local network.

bugOnLine99 wrote:

3) Does MKE have any detailed documentation on how to use all the other Ethernet commands?

For the time being you should use our help files as the source of information regarding any of our libraries.

bugOnLine99 wrote:

4) I notice the following XML code in the "Ethernet_Demo_events_code.c" file:
If I wanted to create my own web page can I modify/re-code the XML code above?

Code that you snipped is not XML, it's HTML in combination with little bit of JavaScript.

bugOnLine99 wrote:

5) Can I replace the code in the "Ethernet_UserUDP()" and the "Ethernet_UserTCP()" functions with my own code in order to create some of my

own control logic.

Of course you can. The aim of this example project is just to give you the basic idea, the rest is up to you.

bugOnLine99 wrote:

6) Does MKE have any tutorial kits for sale which would explain simple code samples for creating our own web pages in the MCU?

Since there is large number of user showing interest in Ethernet example, your suggestion sounds really good. We will definitely consider writing such tutorial.

bugOnLine99 wrote:

7) I would like to create my own web pages along with discrete and analog/numerical control. For example, one can set the heating set point temperature from the web page. What documentation/kits/books/articles would you suggest I read to acquire more knowledge with Ethernet technology for embedded systems.

I would suggest that you try to learn basic syntax of HTML and JavaScript (w3schools.com is good starting point) and then try to figure out how this example is working 'inside out'. That should be enough for you to make your custom web server project.

bugOnLine99 wrote:

I was wondering, does the webserver in the multimedia board require the DHCP to be enabled. Where is it enabled in your Ethernet sample?

There is no need for DHCP to assign IP address when you set it manually.

Best regards.

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bugOnLine99

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 08 Aug 2012 17:15

offline

Joined: 06 Sep 2009
02:09
Posts: 106

Hi dejan.odabasic,

>First of all I'm sorry I didn't mention that IP address is set to 192.168.1.60.

That's okay... no need at all to apologize! 😊

>Also I'll try my best to answer all of your questions.

You have done very well and I really have no problems with your support. You are prompt, clear and to the point. Since I started with MKE, yourself, Filip and Janko have been very helpful and always came through for me. You guys are great.

>>bugOnLine99 wrote:

>>2) In the google chrome browser I enter the following address to access the web server on the multimedia board:

>>"192.168.1.60"

>Would this also work from a smart phone? Ex: iPhone...

>Yes it will if you connect your phone over WiFi to your local network.

This is great... I got this to work this morning. But suppose I am at the country place or downtown and I want to access my web server where my WiFi is not existent. Is there a way to access the multimedia web server page from my iPhone through my ISP via my DSL modem? ?

bugOnLine99 wrote:

>>4) I notice the following XML code in the "Ethernet_Demo_ebents_code.c" file:

>>If I wanted to create my own web page can I modify/re-code the XML code above?

>Code that you snipped is not XML, it's HTML in combination with little bit of JavaScript.

Okay, that's fair enough, but to my surprise I found this code in "Ethernet_Demo.cfg" file:

```
<?xml version="1.0"?>
<MCU_DEVICE_FLAGS>
<DEVICE>
<DEVICE_NAME>P32MX795F512L</DEVICE_NAME>
<VALUE>
<COUNT>4</COUNT>
<VALUE0>
<VAL>$1FC02FF0:$C4070000</VAL>
</VALUE0>
<VALUE1>
<VAL>$1FC02FF4:$0151</VAL>
</VALUE1>
<VALUE2>
<VAL>$1FC02FF8:$14C503</VAL>
</VALUE2>
<VALUE3>
<VAL>$1FC02FFC:$110FF00F</VAL>
</VALUE3>
</VALUE>
```

```
</DEVICE>
</MCU_DEVICE_FLAGS>
```

Isn't this XML and what is this used for ? 😊

>I would suggest that you try to learn basic syntax of HTML and JavaScript (w3cschools.com is good starting point) and then try to figure out how this example is >working 'inside out'. That should be enough for you to make you custom web server project.
Okay that's what I will do... However, is MKE support knowledgeable in case I get stuck... ex: HTML code syntax should work but does not do what it is supposed to do once compiled and downloaded in the multimedia board ?

Okay here are a few last questions... 🤖

Also, there is one function in particular where I don't understand its parameter list and it is:

```
void Ethernet_Enable(unsigned char enFlt);
```

What is MAC Broadcast traffic.... filter flags ... multicast... unicast??? Excuse my ignorance 😞

I see often in your Ethernet help file's description you start off by saying:

"This is MAC module routine."

"This is UDP module routine"

"This is ARP module routine"

"This is DNS module routine"

What do you mean by these exactly 😊

>There is no need for DHCP to assign IP address when you set it manually.
So where exactly is this set manually. Can you show me the line in your Ethernet demo where you call:

```
Ethernet_initDHCP();
```

Also, when to use and not to use:

```
Ethernet_Set_Default_PHY
```

```
Ethernet_Enable
```

```
Ethernet_Disable
```

```
Ethernet_initDHCP
```

Thanks for your help. I really feel like I am asking too many questions 😊
Sorry to be such a bother.

best and sincere regards
Roberto

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dejan.odabasic

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 09 Aug 2012 10:21

 offline

Hello Roberto,

mikroElektronika team

Thank you for you kind words. 😊



bugOnLine99 wrote:

. But suppose I am at the country place or downtown and I want to access my web server where my WiFi is not existent. Is there a way to access the multimedia web server page from my iPhone through my ISP via my DSL modem? 🤔

You need static IP address for you to be able to access to your web server from internet space.
Just like any other web server.

If you have static IP address assigned to your router by your internet provider. It's possible to set up port forwarding and that way make you web server publicly available.

bugOnLine99 wrote:

So where exactly is this set manually. Can you show me the line in your Ethernet demo where you call:

```
Ethernet_initDHCP();
```

IP addresses (network setup) in this example is hardcoded so there is no need for MCU server to ask DHCP server for address.
When you hardcode the IP address you got to know that there is no host which uses the given IP address.

bugOnLine99 wrote:

Okay, that's fair enough, but to my surprise I found this code in "Ethernet_Demo.cfg" file:

Yes, that is XML code and it's used to save microcontroller configuration words, set in edit project window.

Joined: 30 Apr 2012 14:20
Posts: 2416

All our projects use XML files to save config words.

bugOnLine99 wrote:

However, is MKE support knowledgeable in case I get stuck... ex: HTML code syntax should work but does not do what it is supposed to do once compiled and downloaded in the multimedia board ?

We'll do our best as always. 😊

It's important that you take care about string termination character, also note that our C compiler supports escape character. My suggestion is that you first try your HTML code on PC web browser and then try to split it in compiler string constants.

Regarding the last group of your questions I would suggest that you continue studying the Ethernet protocol.

Best regards

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bugOnLine99

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 10 Aug 2012 16:34

offline

Joined: 06 Sep 2009
02:09
Posts: 106

Hello dejan.odabasic,

Quote:

You need static IP address for you to be able to access to your web server from internet space.
Just like any other web server.

Yes... I got this to work as well! I couldn't believe it, I was able to access the web server on the multimedia board from my iPhone.

I did do lots of reading about this though... and I am very happy I learned something 😊

So this is the way I did it:

Initially, in Safari browser, I tried to type, 192.168.1.60 and obviously I was not able to access the MKE web server on the multimedia board 😞

So reading up on tons of articles on the web I proceeded to do the following:

1) In PORT RANGE FORWARDING page of my linksys router, I set a port range. Here's how I did it. In the 'application' field I entered 'xxx', I then assigned port 80 to IP: 192.168.1.60... and made sure I checked off the enable check box. I tried to enter '192.168.1.60' in my safari browser, but this still didn't work.

So I proceeded to step #2...

2) I then "had to" set "PORT TRIGGERING" ! In the port triggering page of my router, at the 'application' field I entered 'xxx', I then assigned the triggered and forwarding ranges to port 80 and made sure I checked off the enable check box.

This still didn't work. So I proceeded to step #3

3) In summary, the MKE multimedia board is connected to my router. My router is then connected to my DSL modem. Therefore I realized that I would need the IP address of my modem if I was to get my iPhone to see my network at all. However, I have a speed stream 5360. This is an old modem and does not have an internal GUI which means I couldn't do any IP routing. Therefore, my ISP sees my router and not my dsl modem. The modem in "my case" is totally transparent. I realized that my router was DHCP enabled which meant that it accepted a dynamic address directly from my ISP. Therefore, I went on "www.whatismyip.com" and retrieved the current IP address assigned to my router from the ISP at this current moment which was '70.52.57.92'. I took this IP and entered it in safari browser and I was now able to connect to the MKE multimedia board 😊

However after a short period of time, the connection failed 😞 So I proceeded to step #4.

4) I realized that the Ethernet demo has a refresh HTML routine. I changed the address from '192.168.1.60' to '70.52.57.92' at the following line:

Code: Select all

```
const code char *indexPath = // Change the IP address of the page to be refreshed
"<meta http-equiv=\"refresh\" content=\"3;url=http://70.52.56.92\">\n"
...
```

and I was finally able to have a solid connection from my iPhone to my web server on the multimedia board 😊

However..... the next day, when I tried it, it didn't work. It was obvious that the ISP's dynamic address changed. So I had to repeat the process and re-assign the new IP address obtained from "www.whatismyip.com" 😞

Now, from what I have read, if I want the address obtained by my ISP to be steady, I have to purchase a static IP as you said 😊

And this costs anywhere from 10\$ to 50\$ a month depending on your location and ISP. 😞

So my question is.... isn't there a way to obtain the router's temporary ip address remotely. Something like... from my iPhone I send an email to my office and this email is able to respond with the current router's dynamic address back to me so I can connect to the MKE multimedia board web server using the current ISP's dynamic IP address ?

I am aware that this is probably out of MKE's support scope but in case you or anyone else has a simple and quick resolution to this... I would really be curious 😊

Now here is another problem I have which *might* be related to MKE support... Not a big deal if there is no resolution though... it's just something weird... here it is:

As I am connected presently and as you know, the multimedia board is connected to my router. My router only has 4 ports. So in reality, one of the ports on my router is used to connect a switch in order to expand the number of ports for my LAN network. So far for the tests I have been discussing in this thread I temporarily disconnected my switch from my router and replaced it with the MKE multimedia board. Now, I would like to do the same thing but this time connecting the MKE multimedia board to my switch instead of my router. I tried to connect the multimedia board to one of the ports on the switch while keeping the same configuration as described above and this does not work. This is weird since a switch is supposed to be transparent to the router and all the network nodes. I have posted this in forums and fellows are saying that this should work just as it worked as when I was connected to the router 😞

Just wondering if you or anyone else has any ideas as to why it wouldn't connect via a switch ?? I must note that everything else on this switch fine 😊

Quote:

Yes, that is XML code and it's used to save microcontroller configuration words, set in edit project window.
All our projects use XML files to save config words.

So I would not have anything to do with this ... right?

Quote:

Regarding the last group of your questions I would suggest that you continue studying the Ethernet protocol.

you bet 😊

Thanks dejan.odabasic! 😊

Regards
Roberto

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dejan.odabasic

Post subject: Re: DTE/DCE RJ45 cables?

Posted: 13 Aug 2012 07:32

offline

mikroElektronika team



Joined: 30 Apr 2012 14:20
Posts: 2416

Hello,

all you written regarding your problem with switch is correct.
There is no reason why MMB32MX7 board shouldn't work when it's connected to switch.

bugOnLine99 wrote:

So I would not have anything to do with this ... right?

There is no need for you to manually edit this file.

bugOnLine99 wrote:

... isn't there a way to obtain the router's temporary ip address remotely.

If you find out how to achieve this, please let us know.

Best regards.

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